INTRODUCTION TO INFORMATION SECURITY

WEEK 4

Caesar Ciphers

An Introduction to Cryptography

What is a Caesar Cipher?

- Caesar used to *encrypt* his messages using a very simple algorithm, which could be easily *decrypted* if you know the key.
- He would take each letter of the alphabet and replace it with a letter at a certain distance away from that letter. When he got to the end, he would wrap back around to the beginning.
- Example with a shift of 3:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

But I Don't Want to Make a Table!

- That's fine! If you know what the shift is, you can use something called *modulus*, which is commonly shortened to mod.
- Let's say we wanted to find 8 mod 5. We would divide 8 by 5 and find the remainder. So, in this case, 8 mod 5 = 3.
- In this case, 5 is called the *modulus*.
- Can you solve these?

19 mod 5	2 mod 5	25 mod 5
4	2	0

How Is That Helpful?

- When using a Caesar cipher, you assign each letter to an index starting from 0.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

- You would then compute the following.

(plain letter index + key) mod (total number of letters)

- This will give you the index of the encrypted letter!
- As you can see, the modulus is the total number of letters in the alphabet. For English, this modulus is 26.

- Mathematically, map letters to numbers:

  a, b, c, ..., x, y, z

  0, 1, 2, ..., 23, 24, 25

- Then the general Caesar cipher is:

  $c = E_K(p) = (p + k) \bmod 26$

  $p = D_K(c) = (c - k) \bmod 26$

- Can be generalized with any alphabet.

Features of Caesar Ciphers

- Stream
- Substitution
- Monoalphabetic

A Simple Example

- Let's say we have a 5-letter/alphabets with only the letters A-E
- First, we assign each letter an index, starting from 0.

A	B	C	D	E
0	1	2	3	4

- We then have to choose a key. For this example, we'll use 2.
- Let's try encoding the word BEAD using the formula for the previous slide.
- The index of the letter B is 1. The key is 2. The modulus is 5, since the alphabet is 5 letters.
- Let's use the algorithm: (1+2) = 3. 3 mod 5 = 3. The index of D is 3, so B would become the letter D.
- Using algorithm on each letter, can you encode the full word?

DBCA

Why Does Modding Work?

- The mod accounts for wrapping back around once you reach the end of the alphabet.

A	B	C	D	E
0	1	2	3	4

- When converting the letter E using the key 2, you first add 4 + 2, which gives you 6. As you can see, the index 6 is beyond the scope of this alphabet. When you do 6 mod 5, you get the remainder of 1, which is how much you need to go back and count from the beginning of the alphabet.

What About Decoding?

- To decode, you do the following:

(cipher letter index – key + total number of letters) mod (total number of letters)

- Does anyone know why we can't just do:

(cipher letter index – key) mod (total number of letters)?

  - By adding the total number of letters, you don't have to end up taking the mod of a negative number.

- See if you can decode DBCA to get back to the word BEAD.

Practice Question:

- Convert the given plain text into cipher text using Caesar cipher.

Plain text: B U Z Z

How Would You Program This?

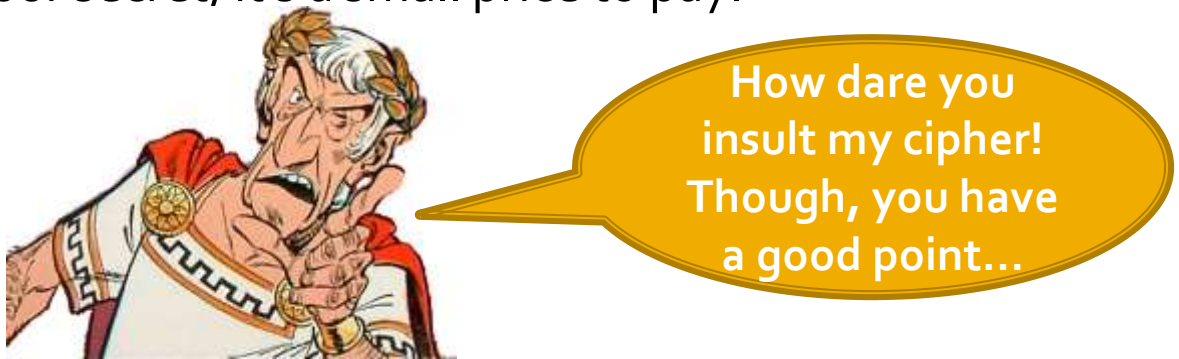
- It's unbelievably simple!
- The mod function is one of the most basic components of a lot of programming languages. You'll learn the mod function for python early on. That means all we have to deal with is indexing the letters.
- Believe it or not, this also is no problem. The solution lies in ASCII.

The Problem with Caesar's Cipher

- It's too easy to break! Let's go back to our five letter alphabet.
- Say you had the word DBCA, but didn't know the key. It would be easy to make a list of all possible keys. If you expected the unencrypted text to be in English, you could easily figure out which word was right:

Shift of 1	**CABE**
Shift of 2	**BEAD ← The clear winner!**
Shift of 3	**ADEC**
Shift of 4	**ECDB**

- If you were using the full English alphabet, this would take longer, but for someone trying to find our your secret, it's a small price to pay!



Monoalphabetic Cipher

- ★ The "cipher" line can be any permutation of the 26 alphabetic characters.
- ★ This would seem to eliminate brute-force techniques for cryptanalysis.
- ★ A single cipher alphabet (mapping from plain alphabet to cipher alphabet) is used per message.
- ★ English language - Nature of plain text is known.

Permutation

- ★ A permutation of a finite set of elements 'S' is an ordered sequence of all the elements of 'S', with each element appearing exactly once.
- ★ For example S = { a , b , c }
- ★ There are six permutations of S.

abc
acb
bac
bca
cab
cba

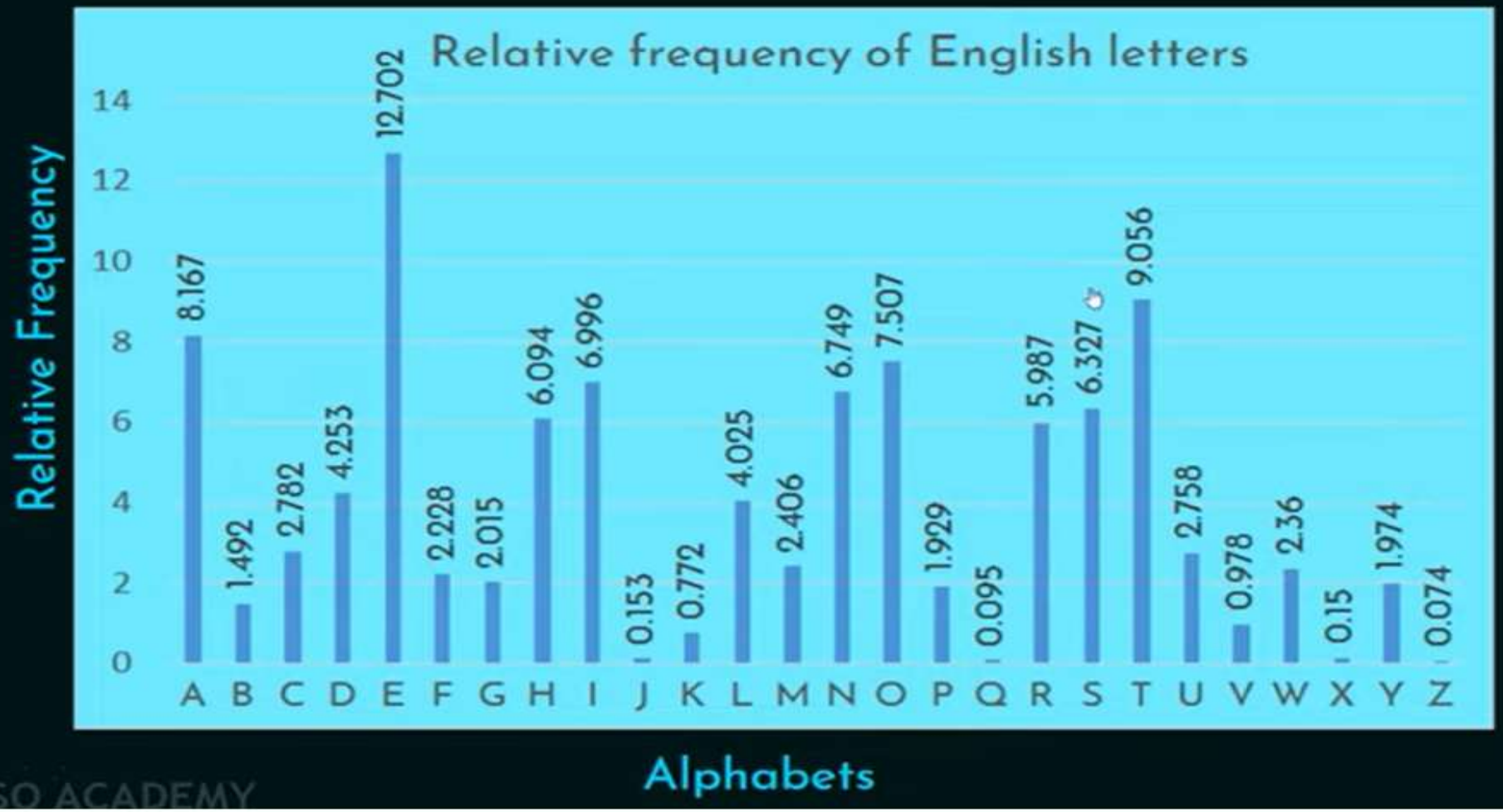
Monoalphabetic Cipher

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Example: Plain Text - Neso

Play (k) ıer Text - DULA

Relative frequency of English letters



ESO ACADEMY

Monoalphabetic Cipher – Example

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

Monoalphabetic Cipher – Example

GZGEWVGRNCP

CT	G	Z	G	E	W	V	G	R	N	C	P
PT	E		E				E				
PT	E		E			T	E				
PT	E		E			T	E			A	
PT	E		E			T	E		L	A	N
PT	E		E			T	E	P	L	A	N

Monoalphabetic Cipher – Example

GZGEWVGRNCP

CT	G	Z	G	E	W	V	G	R	N	C	P
PT	E		E				E				
PT	E		E			T	E				
PT	E		E			T	E			A	
PT	E		E			T	E		L	A	N
PT	E		E			T	E	P	L	A	N
PT	E	X	E	C	U	T	E	P	L	A	N

ESO ACADEMY

Monoalphabetic Cipher – Example

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
t a e e te a that e e a a
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
e t ta t ha e ee a e th t a
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
e e e tat e the t

Monoalphabetic Cipher – Example

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

it was disclosed yesterday that several informal but
direct contacts have been made with political
representatives of the viet cong in moscow

Homework!

Encrypt the plaintext "Attack postponed to tomorrow and do not use our secret paper until further info" using Monoalphabetic cipher technique.

Secret Key: The quick brown fox jumps over the lazy dog.

Note: Ignore the second and latter occurrence of alphabets in the key.